

Charles Darwin, a British Naturalist, went round the world in the ship called *H.M.S. Beagle*, for five years. He published his findings in his book - **ORIGIN OF SPECIES**. According to him evolution involves '**descent with modification**'. With no reservations, one can call him - **THE BIOLOGIST OF THE NINETEENTH CENTURY**.

Unit

1

Diversity of Living World

Biology is the science of life and living processes. The living world comprises an amazing **diversity** of living organisms. However, living organisms show '**Fundamental Similarity**' in many respects. The study of 'Life' involves a multidisciplinary approach, involving many of the principles of **chemistry**, **physics** and **mathematics** too at some places. Biological systems are the most '**complex chemical systems**' on the Earth. The bodies of early organisms are at the 'cellular' level. However large an organism is, '**cells**' of an organism are the '**Basic Units**' of 'structure' and 'function'. Organisms constantly interact with their surrounding environment '**exchanging energy**' and '**recycling minerals**'. The intricate chemical processes of cells are based on the basic principles of chemistry. Every level of biological organisation involves 'energy transactions' governed by the '**Laws of thermodynamics**'. The fundamental source of energy for all biological systems is the 'Sun'. Green plants, over a period of several millions of years, evolved and have mastered the art of converting **solar energy** into **chemical energy** involving complex reactions and pass it on to other higher living organisms to keep **LIFE** going.

Life defies simple definition although one can easily perceive the difference between **inanimate matter** (wind, sea, fire, etc.) and '**living organisms**'. Cellular organisation, ordered complexity, ability to reproduce themselves, showing growth, utilisation and transfer of energy, maintaining constancy in the internal environment (in spite of variations in the surrounding environment), irritability (showing sensitivity to stimuli) and above all, exhibiting the ability to adapt and

evolve (change from simple structure to complex) are some of the '**traits**' of life and living organisms. Diversity in living organisms comes from their ability to 'change'. The process of evolution is continuous and newer forms evolve continuously (**Descent with Modification**), while some perish in the struggle for existence, as envisaged by **Charles Darwin**. Evolution is the 'Fundamental Organising Principle' of life and it answers many questions about life and its complexity - a scientific explanation for 'unity' and 'diversity'. The study and discovery of the molecular structure of DNA by **Watson** and **Crick** (1953) was the major event of the Twentieth Century in the study of biology and it revealed how DNA can **store information** and could serve as the '**chemical basis of inheritance**'. Automatic DNA 'sequencing machines' and the enormous computing power of the present day computers, as a part of the Human Genome Project, have laid inroads to understanding **genetic basis of disease, gene therapy, gene modifications**, etc., to understand life better and to improve the quality of human life, in particular. Comparative studies of DNA, RNA brought in the necessity of classifying living organisms into three **Domains** namely **Bacteria**, **Archaea** and **Eukarya**. The Five Kingdom classification of **Whittaker** followed until a few decades ago, has given place to Six Kingdoms namely - **Bacteria, Archaea, Protista, Fungi, Plantae** (*Metaphyta* or *Viridiplantae*) and **Animalia** (*Metazoa*).

The increase in complexity is right from the simple '**cellular level**' to the '**organism level**' followed by '**populations**', '**communities**' and '**ecosystem**' levels in the **Biosphere** (*all the habitable zones of life on the Earth*). Continuity of life is based on 'heritable' **genetic information** in the **DNA** and in some cases '**RNA**'. In the following subunits of this Unit, you will get an insight into **life**, description of some of the **basic traits of life**, its **complexity of organisation** including **nomenclature** and **classification** from a taxonomical angle.

- 1.1 What is life?
- 1.2 Nature, scope and meaning of zoology
- 1.3 Branches of zoology
- 1.4 Need for classification; zoos as tools for the study of taxonomy
- 1.5 Biological classification
- 1.6 Levels of hierarchy of classification
- 1.7 Nomenclature- Binominal and trinominal
- 1.8 Species concept
- 1.9 Kingdom Animalia
- 1.10 Biodiversity- meaning and distribution, patterns, threats etc.

1.1 WHAT IS LIFE ?

‘**Life**’ is an exclusive property of the various types of living organisms in the world. Recent discovery of fossils of some microscopic organisms extended the known ‘history of life’ on the Earth to about **3.5** billion years. **Biology** is the science dealing with the study of living organisms which are broadly classified into microorganisms, plants and animals. In spite of their wide ‘**diversity**’, they show ‘**fundamental uniformity**’ in possessing *protoplasm* with the genetic substance, the *DNA* which is made up of ‘**nucleotides**’, universally. Living organisms show:

- I. **Cellular organization** and **highly ordered structure**: An organism’s body is made up of one to many ‘**cells**’, which are the **structural** and **functional units** of the body. Each cell is isolated from the cells surrounding it by a limiting membrane the plasma membrane. This membrane controls the exchange of various substances across it. Two major kinds of cells occur in the living world – the **prokaryotic** cells (without distinct nuclei) and the **eukaryotic** cells (with distinct nuclei).
- II. **Complexity of organization**: A fundamental feature of life is the presence of a high degree of ‘order’. Living organisms show hierarchical organization such as cellular level, tissue level, organ level and organ- system level. Several organ-systems make up an **organism**. This type of ‘**ordered complexity**’ is not seen in the non-living things. The hierarchy of life can be schematically represented as follows (starting from the inorganic elements that go into the constitution of protoplasm).



ATOMS → MOLECULES → CELL → TISSUE → ORGAN → ORGAN SYSTEM → ORGANISM → POPULATION → COMMUNITY → ECOSYSTEM → BIOSPHERE.

This representation includes both the inorganic and organic components, as living substance is made up of several inorganic constituents and life processes involve continuous exchange and recycling of inorganic elements.

- III. Sensitivity/ Response to Environment:** It is the property of showing response to external or internal stimuli received through various kinds of 'sense organs'. Living organisms show response to environmental stimuli, which could be physical, chemical or biological. Plants show sensitivity to environmental factors such as light, temperature, water, etc. Ability to show response is called irritability
- IV. Growth:** Growth is one of the *fundamental characters* of living beings (including the unicellular organisms). Increase in mass and increase in number are twin characteristics of growth. Some non-living things also may show tendency of growth but growth in living beings is '**growth from inside**', whereas growth in the non-living things is by accumulation of material *on the surface*. Higher plants show growth all their life producing new branches, leaves, etc., where as it is

'limited' (up to a certain age only) in animals. *Growth cannot be taken as a defining property of living organisms (Ref: NCERT Text Book) as non-living structures also show growth in some respects.*

- V. Energy utilisation/ Energy processing/ Metabolism:** The sum total of all the chemical reactions occurring in the bodies of organisms constitute metabolism and it is a defining feature of all living organisms without exception. Life activities require 'energy' in different forms. The chief source of energy for living organisms is the *sun light*. Energy is transferred from one trophic level to another level. The life processes which build up or store (*conserve*) energy are called **anabolic** processes and the processes involving expenditure of energy are called **catabolic** processes. Together, they constitute - **metabolism**.

ATP is the **chief energy carrier** for various reactions in living systems and it is appropriately described as the 'Cellular Energy Currency'.

- VI. Reproduction:** Living organisms produce young ones of their kind, using molecules of heredity, the DNA molecules (genes), which are passed on to the offspring. It is important to note that '**life comes only from life**' (**biogenesis**) and not from non-living substances. Reproduction, characteristic of all living organisms, occurs by *vegetative*, *asexual* and *sexual* methods. Sexual method involves fusion of gametes forming a zygote, which through various stages of development becomes a young organism. However, some organisms do not reproduce (mules, sterile worker honey bees, infertile human couples, etc.). **Protozoans** reproduce by asexual methods such as 'binary fission'. **Sponges, hydra**, etc., reproduce by 'budding'. In **planarians**, fragmented body (by fission) regenerates the lost parts of the body and becomes a new organism.

- VII. Homeostasis:** Maintenance of relatively constant internal conditions (**steady state**) different from the surrounding environment is called '**homeostasis**'. It is the **dynamic constancy** of the **internal environment** of an organism within a range that the cells can tolerate. Living organisms maintain 'constant internal environment' by various physiological adaptations.

- VIII. Evolution:** Life is not 'static'. It constantly changes (*dynamic*) from simple to more complex forms. Variations in organisms arise through '**mutations**' or '**gene re-combinations**'. Charles Darwin's Natural Selection theory states that living beings accumulate their 'beneficial variations' over a period of time and tend to evolve gradually into new types of organisms (species) after surviving the struggle for existence.

- IX. Senescence and Mortality:** Living beings are born, grow into mature forms, undergo ageing process and finally die. The process of ageing is called 'senescence'. It leads to death / mortality. Living beings are thus '**mortal**'.

All the **life phenomena** like nutrition, respiration, excretion, irritability, etc., are due to **interactions** between the various components of an organism. Properties of tissues are not present in all the constituent cells but arise as a result of interactions among the constituent cells. Similarly properties of cellular organelles are not present in the molecular constituents of the organelle but arise as a result of interactions among the molecular components comprising the organelle. Thus, interactions result in certain properties, at a higher level of organization. This phenomenon is true in the **hierarchy of organizational complexity** at all levels. **Biology is the story of life and evolution of living organisms on the Earth.** All living organisms- the present, past and future are linked to one another by sharing a common **genetic material**, but to varying degrees.

1.2 NATURE, SCOPE AND MEANING OF ZOOLOGY

Biology is a science devoted to the study of living organisms. Science progressed by breaking down complex subjects into their component parts and so today there are numerous branches of biology of which BOTANY, ZOOLOGY and MICROBIOLOGY are the principal, heterogeneous and divergent groups.

Zoology (zoon-animal; logos-knowledge / study) or '**Animal Science**' is a part of the biological science which deals with the study of various aspects of animals, starting from the sponges (Phylum : Porifera) to mammals (Class : Mammalia). The aim of zoology is to explain the **animal world** in terms of **scientific principles**.

Zoology is studied as a 'pure science' (knowledge gaining) and it has application in other branches such as euphenics, eugenics, biotechnology, bioenergetics, bioinformatics, etc. As applied science, it has tremendous scope in agriculture, aquaculture, animal husbandry, human health, diseases, veterinary science, apiculture, sericulture, pharmacology, animal breeding, etc.

1.3 BRANCHES OF ZOOLOGY

Lamarck (1809), a French biologist coined the term '**Biology**', which means the 'study of living organisms'. This diverse science which deals with all aspects of animal life has several sub-branches. A few of them are listed below.

Sub branches of Zoology

1. **Taxonomy** (Taxis - arrangement; nomos - rule, custom)

It is the theory and practice of identification, nomenclature and classification of organisms. The term 'Taxonomy' was coined by A.P. de Candolle.

2. **Morphology** (morphos - form; logos - study)

It deals with the study of form, size, shape, colour and structure of various organisms and their tissues, organs, organ- systems, etc. It includes the following

- i. **External morphology** It is the study of external characters of an organism.
- ii. **Internal morphology** This branch deals with the study of internal structure. It includes the following
 - a) **Anatomy** (ana - up; tome - cutting): It is the study of the internal arrangement of different organs or organ systems in an organism as observed with the naked eye.
 - b) **Histology** (histos - tissue; logos - study): It is the study of microscopic structure of different tissues. This branch is also referred to as '**Microanatomy**'.
3. **Cytology** (Kytos - cells; logos - study):

Cytology deals with the study of form and structure of cells and cell organelles. Cell biology is the branch of science that deals with the study of the cell as a structural and functional unit of living organisms.
4. **Physiology** (Physis - nature of functioning; logos - study):

It is the study of different body functions and processes.
5. **Embryology** (Embryon - embryo; logos - study) and **Developmental biology**

Embryology deals with the study of events that lead to fertilization, cleavages, early growth and differentiation of zygote into an embryo. It is also defined as the branch of biology dealing with the formation and development of embryos. **Developmental biology** is the study of embryonic development and the other developmental processes after birth.
6. **Evolution** (e - out; volva - roll):

It is the study of origin of life and continuous genetic adaptations of organisms to the environment. It also deals with the gradual changes that occur in the living organisms through geological time. Evolution means 'unfolding'. **Herbert Spencer** coined the term '**Organic Evolution**'.
7. **Palaeontology** (Paleo - ancient; on - being; logos - study)

Study of fossilized remains of organisms of the past geological ages is called palaeontology. This includes palaeobotany(fossils of plants) and palaeozoology (fossils of animals).
8. **Ecology** (Oikos - house; logos - study)

It is the study of living organisms in relation to the other living organisms (biotic factors) and abiotic environmental factors surrounding them. **Haeckel** coined the term '**Ecology**'.
9. **Genetics** (Gen - to grow into)

Genetics is the study of inheritance of characters from one generation to the next. It deals with **heredity** and **variations**. **Bateson** coined the term 'Genetics'.

10. Ethology (Gk. **ethos** - character)

The study of the animal behaviour based on the systematic observation, recording, analysis of functions of animals, with special attention to physiological, ecological and evolutionary aspects is called **ethology**.

1.4 NEED FOR CLASSIFICATION

It is impossible to study all living organisms. So, it is necessary to devise some means to make this possible. This process is called '**classification**'. Classification is defined as the process by which anything is grouped into convenient categories based on some easily observable characters. The scientific term used for these categories is 'TAXA' (singular: taxon). Taxa can indicate categories at different levels e.g. Animalia (which includes multicellular animals), chordata, mammalia, etc., represent taxa at different levels.

Hence, based on certain specific characteristics, all the living organisms can be classified into different taxa. This process of classification is called **taxonomy**. Study of external and internal structures, along with the structure of cell, developmental processes and ecological information of organisms are essential and they form the basis of modern taxonomic studies. Hence, **characterisation**, **identification**, **nomenclature** and **classification** are the processes that are basic to taxonomy.

'Zoos' As Tools For Classification

These are the places where wild animals, taken out of their natural habitat, are placed in protected environment under human care (**Ex-situ conservation**). This enables us to learn about the animal's external features, food habits, behaviour (ethology) etc. These observations enable us to systematise the organism and position it in the animal world.

Museums

Museums have BIOLOGICAL SPECIMENS preserved in containers or jars in preservative solutions. Animal specimens may also be preserved as dry specimens. Insects are preserved in insect boxes after collecting, killing and pinning on sheets. Larger animals like birds and mammals are usually 'stuffed' and preserved. Museums often have collections of skeletons of animals too.

1.5 BIOLOGICAL CLASSIFICATION

The living organisms exhibit a great deal of diversity due to variations in their structure and function. So far, over 1.25 million animal species have been identified and described. They show diversity in structure, habits, habitats and modes of life. To understand the interrelationships among the diversified animal groups, a systematic classification is necessary.

History of Biological Classification

Carolus Linnaeus (1707-1788), Father of Taxonomy and Founder of Modern Systematics, introduced the system of hierarchical classification. In the 19th and 20th centuries numerical taxonomy and phylogenetic classification emerged.

Classification

Phylogenetic (Cladistic) Classification

It is an evolutionary classification based on how a common ancestry was shared. Cladistic classification summarises the 'genetic distance' between all species in the 'phylogenetic tree'. In cladistic classification characters such as **analogous characters** (characters shared by a pair of organisms due to convergent evolution e.g. wings in sparrows and patagia (wing like structures) in flying squirrels) and **homologous characters** (characters shared by a pair of organisms, inherited from a common ancestor e.g., wings of sparrows and finches) are followed/taken into consideration. **Ernst Haeckel** introduced the method of representing phylogeny by 'trees' or branching diagrams.

1.6 LEVELS AND HIERARCHY OF CLASSIFICATION

Human beings are not only interested in knowing more about different kinds of organisms and their diversities, but also the relationships among them. This branch of study is referred to as **SYSTEMATICS**. Systematics is the branch of science that deals with the vast diversity of life. It also reveals the trends and evolutionary relationships of different groups of the organisms. These relationships establish the phylogeny of organisms. A key part of systematics is taxonomy. Taxonomic **hierarchy** includes **seven obligate categories** namely **kingdom, phylum, class, order, family, genus** and **species**, and other intermediate categories such as subkingdom, grade, division, subdivision, subphylum, superclass, subclass, superorder, suborder, super family, subfamily, subspecies, etc. If we consider the three Domain classification, we have eight obligate categories.

Linnaeus was the first taxonomist to establish a definite hierarchy of taxonomic categories called taxa (singular: taxon) like kingdom, class, order, genus and species. Haeckel (1888) introduced the taxon **Phylum**. A species sometimes may have more subspecies, which show some morphological variations (intra-specific variations).

Taxonomic Categories: Nowadays the three **Domain** classification is followed. **CARL WOESE** and co-workers observed that many prokaryotes previously classified under 'Prokaryota/ Monera' are more closely related to the 'eukaryotes' and classified them under a separate Domain the **ARCHAEA**. This type of study is called '**MOLECULAR SYSTEMATICS**'.

Now there is a general agreement on the **THREE DOMAIN CLASSIFICATION** of the living organisms namely **DOMAIN-I** : BACTERIA, **DOMAIN-II** : ARCHAEA and **DOMAIN- III**: EUKARYA. (**NOTE: DOMAIN is a taxon higher than 'Kingdom'**).

Bacteria and **Archaea** represent two separate, distinct groups (called Domains). Thus, the modern taxonomists replaced the **Kingdom Monera** with two distinct Kingdoms - **BACTERIA** and **ARCHAEBACTERIA**. Studies on RNAs revealed that *Archaea* and *Eukarya* are more closely related to each other than to Bacteria. According to three domain system there are **six kingdoms** namely Eubacteria, Archaeobacteria, Protista, Plantae, Fungi and Animalia.

1. **Kingdom:** All multicellular, non-saprobic, heterotrophs are included in the kingdom **Animalia**/ Metazoa.
2. **Phylum:** It includes one or more classes. E.g. Phylum Chordata includes the classes Cyclostomata, Chondrichthyes, Osteichthyes, Amphibia, Reptilia, Aves and Mammalia, along with the protochordates. All these are based on common features such as presence of notochord, dorsal hollow nerve cord, pharyngeal slits, post-anal tail, etc., in some stage of the life history.
3. **Class:** It includes one or more related orders. E.g. The class Mammalia includes the orders Rodentia (**rats**), Chiroptera (**bats**), Cetacea (**whales**), Carnivora (**dogs**), Primates (**monkeys** and **apes-gorilla, gibbon** and **man**), etc.
4. **Order:** It includes an assemblage of one or more related families. E.g. The families Felidae and Canidae are included in the order **Carnivora** along with Hyaenidae (**hyenas**), Ursidae (**bears**), etc.
5. **Family:** It includes one or more related genera and can be distinguished from the other families by important characteristic differences. Family felidae includes the genus of cat (*Felis*), genus of leopard (*Panthera*), etc. The members of Felidae can be distinguished from those of Canidae (foxes, dogs, wolves). Adding the suffix-idae to the type generic name forms the name of the family e.g. adding the suffix 'idae' to the type Generic name '**Homo**' gives the name of the Family to which man belongs - **Hominidae**. The name of the subfamily can be coined by adding the suffix-inae to the type generic name (e.g. **Homininae**) and the name of the superfamily by adding the suffix-oidea (e.g. **Hominoidea**).
6. **Genus:** It is a group of related species, resembling one another in certain characters e.g. *Panthera leo* (lion), *Panthera tigris* (tiger) and *Panthera pardus* (leopard) belong to the genus *Panthera*.
7. **Species and Subspecies:** Species is the **basic unit** of classification in the hierarchical taxonomic system. Species is a group of similar organisms sharing a '**common gene pool**' and interbreeding freely, producing 'fertile' offspring. A species occurs in the form of many interbreeding groups called 'populations'.

A species may include **subspecies**. Subspecies is/may be a geographically isolated population of a species, which shows some minor **variations** from the parent population, but are capable of interbreeding with the individuals of other subspecies of the same species. *Subspecies are probably new species in the making.*

For example, the scientific name of crow is *Corvus splendens*. Geographically the crows present in India, Pakistan, Myanmar and Sri Lanka are isolated and evolved into different subspecies- *Corvus splendens splendens* is the **subspecies** in **India and Pakistan**, *Corvus splendens insolens* is the subspecies in **Myanmar** and *Corvus splendens protegatus* is the **subspecies** in **Sri Lanka**. In this system of nomenclature the first word refers to the 'genus', the second to the 'species' and the third to the 'subspecies'.

1.7 NOMENCLATURE

*It is estimated that the number of species known and described ranges between 1.7 to 1.8 million. They are called by their local names (in regional languages), which vary from place to place and even within the same country. So there is need to standardise the naming pattern of living organisms such that a particular organism is known by the same name all over the world/ universally. This process of naming of animals with a distinctive (scientific) names is called **nomenclature**. Naming of organisms is done as per the guide lines of the **International Code of Zoological Nomenclature (ICZN)**. **Binominal nomenclature** (originally called Binomial nomenclature) is used in naming organisms all over the world.*

I. Binominal Nomenclature

Carolus Linnaeus, a Swedish botanist, popularised the '**binomial nomenclature**' by using it in the 10th edition of his book **Systema Naturae**. It is the type of nomenclature in which each organism is provided with an appropriate scientific name consisting of two components, the "**binomen**". The first word refers to the '**genus**' (pl : genera) and the second word is the 'specific epithet'^[1] (species name) . The word that refers to the 'genus' is a '**noun**', and the specific epithet that refers to the 'species' is mostly an '**adjective**'. The generic name begins with a capital letter and the specific name with a small letter. Names must be in Latin or latinised form and are usually '**printed**' in italic type. When '**written**' the two words are to be underlined separately. Let us take the example of a 'lion' to understand binominal nomenclature. The scientific name of lion is ***Felis leo***. In this name, the word '**Felis**' represents the genus, while the word '**leo**', is the specific epithet. The name of the taxonomist follows the scientific name either in full form or in an abbreviated form e.g. ***Felis leo Linnaeus*** or ***Felis leo Linn.***

^[1] Any word or phrase applied to a person or thing to describe an actual or attributed quality(adjective).

or ***Felis leo* L.** It indicates that this species was first described by Linnaeus. *The year of the discovery is written after the name of the person who discovered it - e.g. ***Felis leo Linnaeus, 1758***. When the name of the genus is not the one under which the original author placed a species, or if the generic name is changed subsequently, the original author's name and the year are kept in parenthesis e.g. ***Panthera leo (Linnaeus, 1758)***. It is written so, to understand that Linnaeus originally placed the species name 'leo' under the genus 'Felis' and it was later shifted to the genus Panthera.*

II. Trinominal Nomenclature

It is the extension of the binominal system of nomenclature. This system permits the designation of subspecies with a three-worded name called '**trinomen**'. This system of nomenclature is used to name the subspecies, which is a category below the level of species. The word denoting the name of sub species also begins with a small letter and it is a latinised word, printed in italics e.g. ***Homo sapiens sapiens***; ***Corvus splendens splendens***.

TAUTONYMY: The practice of naming the animals, in which the generic name and species name are the same, is called tautonymy. So the name is called tautonym e.g. ***Axis axis*** (spotted deer); ***Naja naja*** (the Indian cobra).

1.8 SPECIES CONCEPT

I. Species

Species is the 'basic unit' of classification. Species is a Latin word meaning 'kind' or 'appearance'. John Ray in his book 'Historia Generalis Plantarum', used the term 'species' and described it on the basis of common descent (origin from common ancestors) as a group of morphologically similar organisms. Linnaeus considered species, in his book 'Systema Naturae', as the basic unit of classification. Buffon, in his book 'Natural History', proposed the idea of evolution of species which is the foundation for the biological concept of evolution. This biological concept of species (dynamic nature of species) became more popular with the publication of the book "The Origin of Species" by Charles Darwin.

Buffon's biological concept of species explains that species is an interbreeding group of similar individuals sharing the common 'gene pool', and producing fertile offspring. Species is considered a group of individuals which are:

1. Reproductively isolated from the individuals of other species – **a breeding unit.**
2. Sharing the same ecological niche – **an ecological unit.**
3. Showing similarity in the karyotype – **a genetic unit.**
4. Having similar structure and functional characteristics – **an evolutionary unit.**

Let us consider the following examples

Example- 1: *Apis indica*, *Apis dorsata*, *Apis mellifera* and *Apis florea*

In the above example '*indica*', '*dorsata*', '*mellifera*' and '*florea*' are different species belonging to the same genus called *Apis*.

Example- 2: *Pheretima posthuma*, *Periplaneta americana* and *Panthera leo*. In the example 2 the words '*posthuma*', '*americana*' and '*leo*' are names of different species belonging to different genera.

*Sometimes closely related species of a genus can interbreed, but they generally give rise to sterile offspring. A cross between a female donkey and a male horse gives rise to the sterile offspring called '*Hinny*' (a sterile hybrid).*

Dobzhansky introduced the concept of '**Mendelian Population**' while defining a species. A Mendelian population is a group of sexually reproducing individuals within which mating takes place. They share a 'common gene pool'. The members of a species show **assortative** (*preferential*) mating. Populations of a species inhabiting different geographical areas are in a continuous process of adapting to the conditions of their surrounding environments. This leads to the evolution of new species, in course of time. Thus, species is **dynamic**.

Find out the scientific names of at least five animals around your home.

1.9 KINGDOM: ANIMALIA

Animalia includes eukaryotic, multicellular heterotrophs (generally obtain nourishment by ingestion i.e, holozoic method and digesting the ingested food). They have specific body plans and do not possess cell walls and photosynthetic pigments. Tissue formation is common except in the sponges. Higher forms show elaborate sensory and neuromotor mechanisms. Organ and organ-systems are developed in most of the groups. Nutrition is **holozoic** and a few are parasitic. Reserve food is mostly in the form of **glycogen**. These are mostly motile and move in search of food (except the sedentary forms). Muscle cells, sensory cells and nerve cells are present. Reproduction is mostly by sexual method; some reproduce by asexual



methods. Animalia includes simple sedentary organisms such as sponges to the highly evolved vertebrates, the mammals.

Classification of Kingdom Animalia

Kingdom Animalia is divided into two subkingdoms based on the organization of cells into tissues or higher levels of organisation.

- i) **Parazoa:** These are multicellular animals without the formation of well defined tissues. The only phylum included in this group is Porifera (e.g. *Sponges*).
- ii) **Eumetazoa:** These are multicellular animals with well defined tissues and higher levels of organization such as organs and organ- systems in the body. This group includes two taxonomic levels called Grades.

Grade:1. Radiata or Diploblastica

These are the first true metazoans or eumetazoans with radial symmetry and diploblastic body (having two primary germinal layers, the ectoderm and endoderm) Phyla : Cnidaria (e.g. *Hydra*, *Aurelia*, etc.) and Ctenophora (comb jellies) are included in this Grade.

Grade: 2. Bilateria or Triploblastica

This Grade includes eumetazoans with bilateral symmetry and triploblastic body (having three primary germinal layers, the ectoderm, mesoderm and endoderm). This includes two taxonomic levels called **Divisions**.

Division: I: Protostomia (*Proto* : first ; *Stomium* : mouth)

The eumetazoans in which blastopore develops into mouth are referred to as the protostomians . In the protostomes cleavage pattern is **spiral** and **determinate**. This Division includes three Sub-Divisions.

Sub-division: i) Acoelomata: (*a*:without; *coelom*: bodycavity; *ata*:having)

These organisms are the first triploblastic eumetazoans with **solid body plan** (without a body cavity/space between the body wall and visceral organs). The perivisceral space is filled with a tissue called parenchyma/ mesenchyme (derived from the germinal layer called mesoderm) e.g. Phylum Platyhelminthes (flat worms).

Sub-division: ii) Pseudocoelomata (*pseudo*:false; *coelom*:bodycavity; *ata*:having):

Animals of this group possess a body cavity between the body wall and the alimentary canal (perivisceral space derived from the embryonic blastocoel).

However it is not a 'true coelom/ secondary body cavity' as it is not lined by mesodermal epithelial layers (parietal and visceral peritoneal layers). Pseudocoel is a remnant of the embryonic blastocoel/ primary body cavity e.g., the phylum Nematoda. (Note: Truecoelom (eucoelom) is called secondary body cavity).

Sub-division: iii) Schizocoelomata: (*schizo*:splitting; *coelom*:body cavity; *ata*:having) They have a ‘true coelom’, which is a ‘schizocoel’. It is formed by **splitting** of the mesoderm into outer somatic and inner splanchnic layers e.g. the phyla Annelida, Arthropoda and Mollusca. (Note: In the vertebrates coelom is formed by the splitting of the mesoderm (called secondarily schizocoelic)).

Division: II) Deuterostomia (*deuteron*: secondary; *stomium*: mouth)

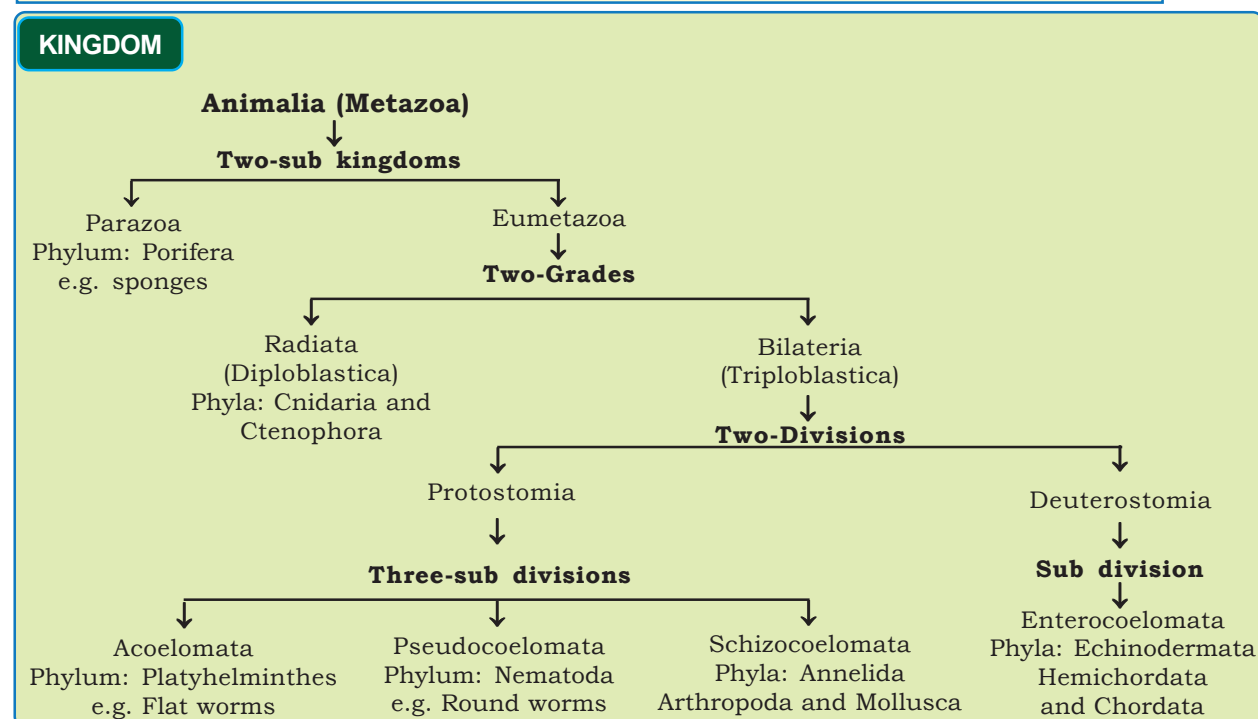
These are eucoelomates in which anus is formed from or near the blastopore. Mouth is formed later (secondarily), away from the blastopore at the opposite end. In deuterostomes cleavage pattern is ‘**radial**’ and ‘**indeterminate**’. It includes only one subdivision, **Enterocoelomata**.

Sub-division: Enterocoelomata

They have a true coelom, which is an ‘enterocoel’. It is formed by the out pouching of the archenteron. The phyla Echinodermata, Hemichordata and Chordata are included in this subdivision.

Schizocoelomates and enterocoelomates are together called ‘eucoelomates’ as they possess a ‘true coelom’.

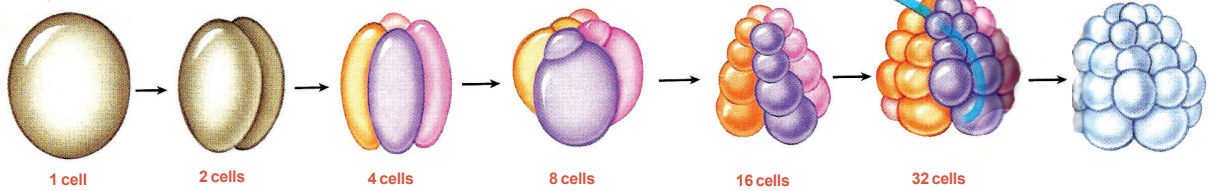
Based on the knowledge gained, can you write the taxonomic position of human being (taxonomic hierarchy) starting from the Kingdom to the species/subspecies? (Take the help of your teacher, if you fail to do).



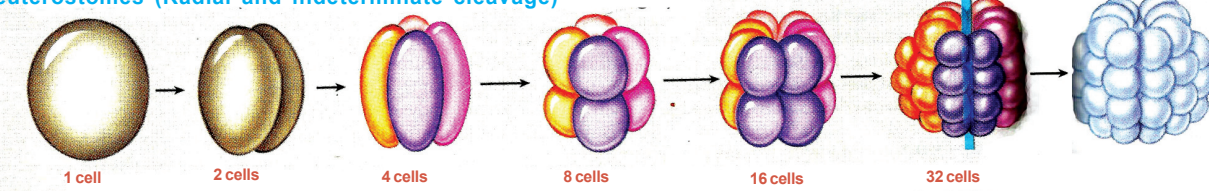
Note : Embryos of protostomes are called ‘**mosaic embryos**’ and those of deuterostomes are called ‘**regulative embryos**’

Diagrams showing embryonic development in protostomes and deuterostomes.

Protostomes (Spiral and determinate cleavage)



Deuterostomes (Radial and indeterminate cleavage)



1.10 BIODIVERSITY

I. What is Biodiversity?

When we observe our surroundings we find different kinds of organisms which vary in size, form, feeding habits, behaviour, etc. For example there are more than 20,000 species of ants, 3,00,000 species of beetles, 28,000 species of fishes and 20,000 species of orchids. This variation of life at various levels of biological organization is termed as biodiversity.



II. Levels Of Biodiversity

Biodiversity exists not only at the species level but at all levels of biological organization ranging from macromolecules within the cells to **biomes** (biotic community in a large area).

The term biodiversity was popularized by the sociobiologist **Edward Wilson** to describe the combined diversity at all levels of biological organization. The three levels of biodiversity are:

1. Genetic diversity
2. Species diversity
3. Ecological diversity.

1. Genetic diversity

It is the diversity of genes within a species. A single species may show high diversity at the genetic level over its distributional

range. For e.g. *Rauwolfia vomitoria*, a medicinal plant growing in the Himalayan ranges shows great genetic variation, which might be in terms of **potency** and **concentration** of the active chemical (**reserpine** extracted from it is used in treating high blood pressure) that the plant produces. India has more than 50,000 different strains of rice, and 1,000 varieties of mangoes. **Genetic diversity** increases with environmental variability and is advantageous for its survival.

2. Species Diversity

It is the diversity at the species level. e.g: amphibian diversity in the Western Ghats is greater than that of the Eastern Ghats.

3. Ecological diversity

Diversity at a higher level of organization, i.e. at the ecosystem level is called 'Ecological diversity'. e.g: India with its deserts, rain forests, mangroves, coral reefs, wet lands, estuaries and alpine meadows has greater ecosystem diversity than many other countries such as the Scandinavian country Norway.

The three indices of ecological diversity are- **Alpha**, **Beta** and **Gamma** diversity

i. Alpha diversity: It is measured by counting the number of taxa (usually species) within a particular area, community or ecosystem.

ii. Beta diversity: It is the species diversity between two adjacent ecosystems and is obtained by comparing the number of taxa **unique to each of the ecosystems**.

iii. Gamma diversity: It is the measure of the **overall diversity for different ecosystems** within an ecological region (**'Ecological Region'** is a large area constituting a natural ecological community with characteristic flora and fauna, bounded by natural boundaries).

III. Other attributes of biodiversity

A species unique to a given area is called **endemic species**. Pattern of biodiversity depends on factors such as i) Latitude and ii) Species-area relationship

1. Latitudinal Gradient in Diversity

Biodiversity is not uniform throughout the world but shows rather uneven distribution. The most important pattern of biodiversity is latitudinal gradient in diversity. This means that there is an increasing diversity from the poles to the equator (terrestrial biodiversity increases from the poles to the equator). There is a vast majority of species concentrated in the tropics and sub tropical regions. This means localities at lower latitudes have more species than localities at higher latitudes.

Tropics harbour more species than temperate or polar areas for e.g. The tropical Amazon rain forest in south America has the greatest biodiversity on the Earth. Species diversity is more in the tropics. The following data explains latitudinal gradient :

Place	Number of species of birds	Latitude
Colombia	1400	0°N
New York	105	41°N
Green land	56	71°N

From the above data it is clearly evident that as the latitude increases the species diversity decreases.

Reasons for greater biodiversity in the tropics:

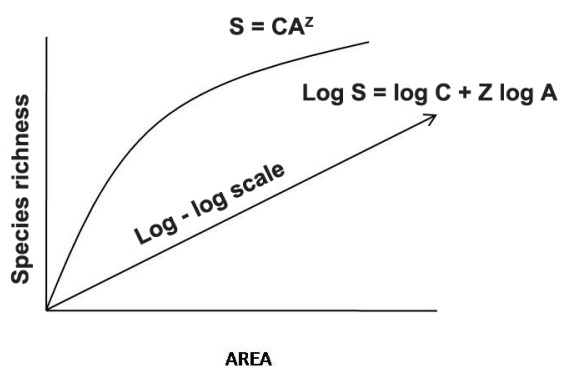
Reason 1: Tropical latitudes have remained relatively undisturbed for millions of years and thus had a long 'evolutionary time'. As long duration was available in this region for speciation, it led to the species diversification. (Note: The temperate regions were subjected to frequent glaciations in the past).

Reason 2: Tropical climates are relatively more constant and predictable than that of the temperate regions. Constant environment promotes **niche specialization** (how an organism responds, behaves with environment and other organisms of its biotic community), and this leads to greater species diversity.

Reason 3: Solar energy, resources like water etc., are available in abundance in this region. This contributed to higher productivity in terms of food production, leading to greater diversity.

2. Species-Area relationships

Before we learn relation between species richness and the area available to them, let us learn the term 'species richness'.



Species Richness is the number of species per unit area. The more the number of species in an area the more is the species richness.

Alexander von Humboldt observed that within a region, species richness increased with increasing explored area, but only up to a limit. In fact, the relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, fresh water fishes) turns out to be a non-linear curve. On a logarithmic scale, the relationship is a straight line described by the equation.

$$\log S = \log C + Z \log A$$

where

S = species richness

A = area

Z = slope of the line (regression coefficient)

C = Y-intercept

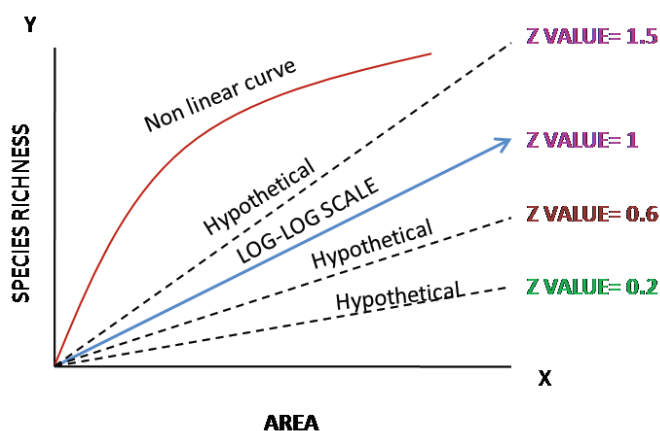
Ecologists have discovered that the value of Z lies in the range of 0.1 to 0.2 regardless of the taxonomic group or the region (whether it is California or New York or Britain). If you analyse species-area relationship among very large areas like entire continents the slope of the line is much steeper (Z values in the range of 0.6 to 1.2) for example for frugivorous (fruit-eating) birds and mammals in the tropical forests of the different continents, the slope is found to be **1.15**.

What do **steeper slopes** mean in this context ?

A simplified explanation to help Biology Teachers

Teachers of Biology might find it difficult to understand and explain the mathematical expression represented in this part. The graphic representation is essentially intended to explain the relationship between 'species richness' and 'area'. In a specified area the relation between ' S ' (species richness) and ' A ' (area) is $S = CA^Z$ and it is represented by a **Non-linear curve**.

For easy understanding, to get a **linear relation** between S and A , '**log scale**' is taken. Logarithmic relation between A and S is given by $\log S = \log C + Z \log A$. The '**curve**' shows that species richness increases with increase in area to a certain extent and approaches an **equilibrium state/ stable state**. It does not indicate the '**specific rate**' at which species richness increases with reference to area. The '**SLOPE**' of the graph drawn between $\log A$ (taken along **X axis**) and $\log S$ (taken along **Y axis**) is Z . Increase in the 'slope' indicates **increase in Z** indicating 'increase in species richness', as shown in the graph. For example, if the log-log scale makes 45° with X axis, Z is **1**. If the slope is less, Z is less than '**1**' and if the slope is more than 45° the value of Z is more than **1**. Now it is easy to understand that the value of Z in a '**tropical rain forest**' is more than 1, as represented by the 'increased slope'.



Importance of Species Diversity to the Ecosystems

Communities with more types of species (more biodiversity) tend to be more stable than those with less number of types of species. Stable communities generally withstand disturbances (natural or man-made). **Tilman's** experiments with 'outdoor plots' showed that 'plots with more species showed less variations in biomass year-to-year'. He also showed that increased **diversity contributed to higher productivity**.

What if we lose a few species? Will it affect man's life? **Paul Ehrlich's** experiments 'the **RIVET POPPER**' hypothesis, taking an aeroplane as an ecosystem, explains how removal of one by one 'rivets' (species of an ecosystem) of various parts can slowly damage the plane(ecosystem)-shows how important a 'species' is in the overall functioning of an ecosystem. Removing a rivet from a seat or some other relatively minor important parts may not damage the plane, but removal of a rivet from a part supporting the wing can result in a crash. Likewise, removal of a '**critical species**' may affect the entire community and thus the entire ecosystem.

IV. Role of biodiversity

Biodiversity is beneficial to human beings as it plays an important role at various levels of development and to explain the role played, the different aspects are categorised into three types,

1. Narrowly utilitarian argument
2. Broadly utilitarian argument
3. Ethical argument

1) Narrowly utilitarian argument: Human beings derive countless economic benefits from nature as biodiversity is a **reservoir of resources**. Food (cereals, pulses & fruits), Firewood, Fibre, Construction material, Industrial products (tannins, lubricants, dyes, resins, perfumes, rubber, latex, cork, etc.) and products of medicinal importance (for example Anticancer drugs – **Vinblastin** from *Vinca rosea*, **Digitalin** from the 'fox glove' plant (*Digitalis purpurea*) to treat certain cardiac problems, etc.) are obtained from diverse living organisms which are economically important. Bio-prospecting nations endowed with rich biodiversity can expect to reap enormous benefits.

2) Broadly utilitarian argument: It explains that biodiversity plays a major role in many ecosystem services that nature provides, for which we cannot put a price tag (*we cannot fix its price*). To exemplify this, consider the following -

e.g.1 : Amazon produces 20 % of the total oxygen in the earth's atmosphere.

e.g.2 : Pollination by pollinating agents(bees, birds, bats, etc.) without which plants do not produce fruits and seeds. There are other benefits also, the ones which cannot be measured in terms of money, like aesthetic pleasure derived by taking a walk in the woods.

- 3) **Ethical argument** : It relates to what we owe to plants, animals and microbe species with which we co-exist on this planet. The moral duty to care and pass on biological legacy to future generations is the need of the hour as every species has an *intrinsic value*.

V. Threats to Biodiversity

The following are the '**four major causes**' (**THE EVIL QUARTET**) for accelerated rates of species extinction in the world.

1. **Habitat Loss And Fragmentation** : These are the most important reasons for the loss of biodiversity.
 - a) Deforestation-leads to species extinction in forests e.g: tropical rain forests once covering 14% of the earth's land surface is now not more than 4%.
 - b) Conversion of forest land to agricultural land e.g: the AMAZON RAIN FOREST, called '**lungs of our planet**', harbouring innumerable species is cut and cleared to cultivate SOYA BEANS or conversion to grass lands for raising beef cattle.
 - c) Pollution enhances degradation of habitats and threatens the survival of many species as pollutants change the quality of the environment.
 - d) Fragmentation of habitat leads to population decline e.g: mammals and birds requiring large territories and certain animals with migratory habits are badly affected.

Fragmentation

It is the process of formation of discontinuities in the natural habitats due to geological processes or human activities. Geological fragmentation may lead to speciation, but fragmentation caused by man (human activity) mostly leads to extinction of many species.

2. **Over-exploitation: When need turns to greed it leads to over exploitation**
e.g., Steller's sea cow (sea cow named after Steller, a naturalist), passenger pigeon (which existed in North America) are extinct due to over-exploitation by humans. The existence of many commercially important marine fishes are endangered as they are over harvested.
3. **Invasion Of Alien Species** : When alien species are introduced into a habitat, they turn **invasive** and establish themselves at the cost of the indigenous species (organisms which occur naturally in a particular region).

E.g.1: **Nile perch** introduced into Lake Victoria, in east Africa led to the extinction of 200 species of cichlid fish in the lake.

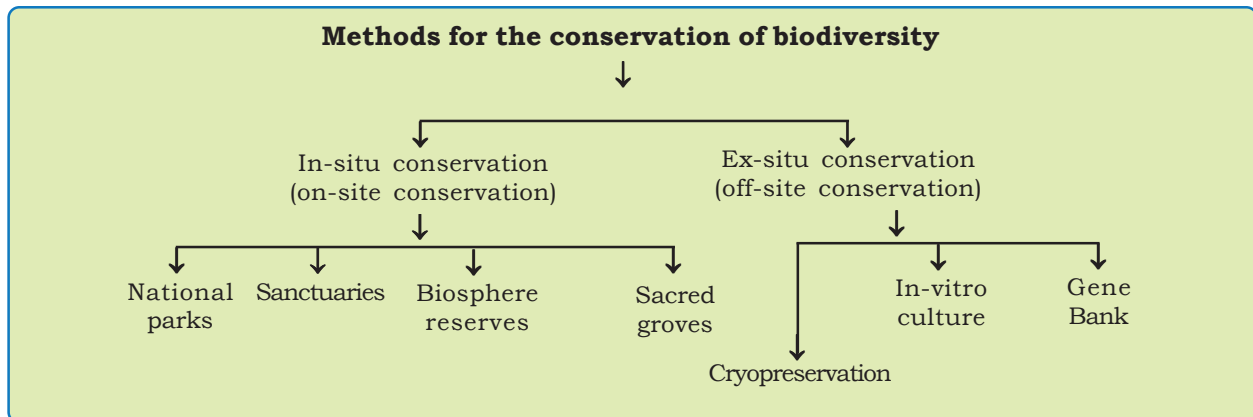
E.g.2: Illegal introduction of exotic **African catfish**, *Clarias gariepinus*, for aquaculture purposes is posing a threat to the indigenous cat fishes.

E.g.3: When exotic and invasive weeds like the ‘carrot grass’ (*Parthenium*), ‘spanish flag’ (*Lantana*), ‘water hyacinth’ (*Eichhornia*) are introduced into our ecosystems they not only damaged the environment but also threatened the very existence of native species.

4. **Co-extinctions:** In an obligate association between a plant and an animal, if a plant becomes extinct, the animal also becomes extinct as seen in a parasite and host association. If the host becomes extinct, parasite meets the same fate. Another association which explains co-extinction is **plant-pollinator mutualism** where extinction of one invariably leads to the extinction of the other.

VI. Methods of Conservation of Biodiversity?

The conservation of biological diversity has become a global concern. There are basically two main types of conservation options-**In-situ** conservation and **Ex-situ** conservation. In-situ is usually seen as the ‘ideal conservation strategy’. Ex-situ conservation can provide a **backup solution** to in-situ conservation projects.



1. In-situ conservation (On-site conservation)

In-situ conservation is the process of protecting an animal species in its natural habitat. The benefit is that it maintains recovering populations in the surrounding where they have developed their distinctive properties. Conservationists identified certain regions by name ‘**Biodiversity hot spots**’ for maximum protection as they are characterized by very high levels of species richness & high degree of endemism. By definition ‘Biodiversity hot spot’ is a ‘Biogeographic Region’ with a significant reservoir of biodiversity that is under threat of extinction from humans. They are Earth’s biologically ‘richest’ and ‘most threatened’ **terrestrial** Ecoregions.

The concept of Biodiversity hot spots was proposed by **Norman Myers**. There are about 34 biodiversity hot spots in the world. As these regions are threatened by destruction, habitat loss is accelerated e.g. :I) Western Ghats and Srilanka ; II) Indo

Burma; III) Himalayas in India. Ecologically unique and biodiversity rich regions are legally protected as in 1. **Biosphere Reserves**–18 (18th is PANNA in Madhya Pradesh), 2. **National Parks**–90, 3. **Sanctuaries**–448.

i. Biosphere Reserves

An area which is set aside, minimally disturbed for the conservation of the resources of the biosphere is '**Biosphere reserve**'. Latest biosphere reserve (17th biosphere reserve in India) is **Seshachalam hills**.

ii. National Parks

A National Park is a natural habitat strictly reserved for protection of natural life. National Parks, across the country, offer a fascinating diversity of terrain, flora and fauna. Some important National Parks in India are - **Jim Corbett National Park** (the first National Park in India located in Uttarakhand), **Kaziranga National Park** (Assam), **Kasu Brahmananda Reddy National Park**, **Mahavir Harina Vanasthali National Park** (AP), **Keoladeo Ghana National Park** (Rajasthan), etc.

iii. Sanctuaries

Specific **endangered faunal species** are well protected in wildlife sanctuaries which permits eco-tourism (as long as animal life is undisturbed). Some important Sanctuaries in India (AP) include-Koringa Sanctuary, Eturnagaram Sanctuary, Papikondalu Sanctuary.

iv. Sacred Groves

1. A smaller group of trees than a forest is called grove.
2. A grove of trees of special religious importance to a particular culture is called sacred grove.
3. In these regions all the trees of wild life are venerated (respected) and given total protection.

The following is a list of Sacred Groves in INDIA

Name	State
Khasi and Jaintia Hills	Meghalaya
Aravalli Hills	Rajasthan and Gujarat
Western Ghat region	Karnataka and Maharashtra
Sarguja, Bastar	Chhattisgarh
Chanda	Madhya Pradesh

In Meghalaya, Sacred Groves are the last refuges for a large number of rare and threatened species.

2. Ex-situ conservation:

In ex-situ conservation threatened animals are taken out of their natural habitat and placed in special settings where they are protected. This includes Zoological Parks. Advancement in ex-situ preservation techniques such as **cryopreservation** are helping us protect endangered species (cryopreservation is the preservation of, for example, gametes, embryos of threatened species, etc., at -196°C). In vitro culture, gene banks are mostly used for plants.

VII. IUCN Red data books

International Union for the Conservation of Nature and Natural Resources (IUCN) is the world's main Authority on the issues of conservation status of species.

All the threatened species are listed in the **Red Data Books** published by the IUCN. These species are classified into different categories based on degree of risk and they are chiefly :

- a) Critically endangered
- b) Endangered
- c) Vulnerable

VIII. Conservation of wild life in INDIA

This is through legislation, preservation and organisations

1. Legislation: Under the provision of the **Wildlife Act of 1972**, killing endangered wild animals is strictly prohibited. Trading wildlife products (like tusks, rhino's horns, etc.) is a punishable offence.

2. Preservation: National Parks, Sanctuaries, Biosphere Reserves, Sacred Groves etc. are different regions which are earmarked to protect diverse fauna and flora.

3. Organisational Protection: Organisations which are set up to prevent destruction of India's wild life are

1. Wild life protection society of India (**Dehradun**)
2. Zoological Survey of India (**Kolkata**)

Conservation of biodiversity is a **global necessity**. It is the collective responsibility of all nations to protect the diverse living forms on the planet. One such step was in the form of **EARTH SUMMIT** (1992-Rio de Janeiro) and the other being **WORLD SUMMIT** on sustainable development (Johannesburg-South Africa). They focussed on significant reduction in the current rate of loss of biodiversity at the global, regional and local levels. Efforts must be intensified to pass on our biological legacy to the future generations.

IX. Threatened Species in India

The Asiatic lion	<i>Panthera leo persica</i>
The black buck	<i>Antelope cervicapra</i>
Red panda	<i>Ailurus ochraceus</i>
The lion-tailed macaque	<i>Macaca silenus</i>
Tiger	<i>Panthera tigris</i>
Kashmiri stag	<i>Cervus elaphus hanglu</i>
Elephant	<i>Elephas maximus indicus</i>
Pygmy hog	<i>Sus salvanius</i>
Siberian crane	<i>Grus leucogeranus</i>
Slender loris	<i>Loris tardigradus</i>

In conclusion-Nature is a repository of diverse life. Intrusion into nature's domain distorts its equilibrium. Preserving nature is a collective 'Global Responsibility'. Man should conserve and protect nature in his own interest and for the future generations.

GLOSSARY

Nutrition: The process by which an organism assimilates food and uses it for growth and maintenance.

Protein: Building blocks of life which are polymer chains made of amino acids linked together by peptide bonds.

Prokaryotes: Single celled organisms that lack a membrane bound nucleus.

Eukaryotes: Organisms whose cells contain a membrane bound nucleus and other complex membrane bound cell organelles.

Glycogen: It is a polysaccharide that is the principal storage of glucose in animal cells analogous to starch in plants.

Chordata: A phylum which includes animals showing notochord at some stage in life cycle.

Phylogeny: The evolutionary history of an organism.

Analogous Characters: They denote similarity in function without necessary anatomical similarity (wings of birds and butterflies).

Assortative mating: It is a selective mating/ non-random mating within their population.

Tissue: A group of cells of similar origin that perform a similar function.

Holozoic: It is a method of nutrition that involves ingestion of liquid or solid organic material.

Ctenophora: An invertebrate phylum of animals which are commonly called 'comb jellies'.

Spiral cleavage: It is characteristic of the protostomes in which planes of cell division are diagonal to the vertical axis/polar axis of the embryo.

VERY SHORT ANSWER TYPE QUESTIONS

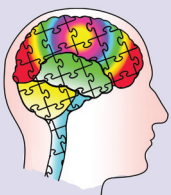
1. Define the term metabolism. Give any one example.
2. How do you differentiate between growth in a living organism and non-living object?
3. What is biogenesis?
4. Define the term histology. What is it otherwise called?
5. Distinguish between embryology and ethology.
6. In a given area, remains of animal that lived in the remote past are excavated for study. Which branch of science is it called?
7. 'Zoos are tools for classification' Explain.
8. Where and how do we preserve skeletons of animals, dry specimens etc?
9. What is trinomial nomenclature? Give an example.
10. What is meant by tautonymy? Give two examples.
11. Differentiate between Protostomia and Deuterostomia.
12. 'Echinoderms are enterocoelomates' Comment.
13. What does ICZN stand for?
14. Give the names of any four protostomian phyla.
15. What is ecological diversity? Mention the different types of ecological diversities.
16. Define species richness.
17. Mention any two products of medicinal importance obtained from Nature.

18. Invasion of an Alien species leads to extinction of native species. Justify this with two examples.
19. List out any four sacred groves in India.
20. Write the full form of IUCN. In which book threatened species are enlisted.

SHORT ANSWER TYPE QUESTIONS

1. Explain the phylogenetic system of biological classification.
2. Explain the hierarchy of classification.
3. What is meant by classification? Explain the need for classification.

4. Define species. Explain the various aspects of 'species'.
5. What is genetic diversity and what are the different types of genetic diversity?
6. What are the reasons for greater biodiversity in the tropics?
7. What is the 'evil quartet'?
8. Explain in brief 'Biodiversity Hot Spots'.
9. Explain '**Rivet Popper**' hypothesis.
10. Write short notes on In-situ conservation.



For ignited minds

- Not For Evaluation

1. If you drink a 'sugar rich' soft drink , the sugar levels in your blood increase quickly. If the sugar levels in blood increase sensitive cells such as 'brain cells' are damaged. Your pancreatic islets produce a hormone called 'insulin' which allows uptake of sugars by body cells so that the body cells utilise them and generate ATP for cellular life activities. The sugar levels of your body fluids come down to normal 'tolerable levels'. What do you think is the 'biological phenomenon' that happened in this case?

Clue: The body always tries to maintain equilibrium state.

2. What, in your expectation, will happen when 'cancer cells' are starved?

Answer: When cancer cells are starved, they cannot show growth .If there is no growth, cells do not multiply. Anabolism is a necessary part of life.

3. What, in your view are reasons for the occurrence of 'variations' in living organisms and the role played by such variations in nature?
4. Why, in your view, children are not 'exact copies' of their parents, when they do inherit parental DNA, which governs their body features?
5. On a cold winter day when you walk outside , you tend to 'shiver'. Can you explain why such 'shivering' is caused?
6. You have learnt about ageing (senescence) in your lesson. Would you like to know one more interesting concept about ageing - It is also controlled by **genes** or to put it in other words **ageing** is also '**genetically**

programmed '. Have you heard of '**Dolly**' the lamb produced by '**cloning**' of the 'udder cell' (somatic cell of the mammary glands) of **Finn Dorset Ewe** (Finn Dorset is a variety of sheep in England ; Ewe is a 'Female sheep'). The average life span of Finn Dorset sheep is about **12** years by which time it develops 'senescence' and dies. 'Dolly' was developed **asexually** by **cloning**, like how a single zygote cell develops into a young one in the case of sexual reproduction. **Do you know how long 'Dolly' lived ?** It is just about **six years**.

Can you guess why it lived for only **six years**, when the average life span of that type of sheep is **twelve** years? *The answer is-*'Dolly', when born, was already **six years old**, '**genetically speaking**', as it developed from the udder cell of a **six year** old sheep, the Finn Dorset ewe. When Dolly was 6 years old, its genetic material is 12 years old, hence the death .

7. Do you agree with the statement the log-log scale linear relation in the south American tropical forests will be $> 45^\circ$?
8. When the human zygote undergoes the first division into two cells, if the two cells are separated they develop into identical twins. Will the same thing happen in the case of earthworm? Why/ Why not?
9. In your opinion can a population be made up of a group of intra-specific variants'? if your answer is Yes, how do you defend?
10. Which among the following encompasses the others - Genus / Grade / Division / Class / Order?